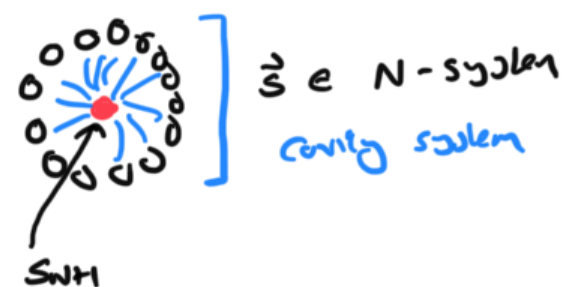


Key outline of cavity method

- ① take a system of size N with config $\vec{S}$
 - ② add a new $N+1$ spin S_{N+1}
and couple it to N -spin
system with new coupling
constants
 - ③ S_{N+1} couples to $\vec{S}$ through
some single function of $\vec{S}$
ie $h_N(\vec{S})$ is a cavity field
 - ④ $h_N(\vec{S})$ has some assumed statistics
in the N -spin system
 - ⑤ write down a distribution for
 $P(S_{N+1}, h_N)$
 - ⑥ compute statistics of S_{N+1} by marginalizing out h_N
 - ⑦ Make sure statistics of S_{N+1}
is consistent with statistics of $h_N(\vec{S})$
since they are both part of the same
 $N+1$ system
- this leads to self consistent equations for
the statistics of the system



A simple example: the ferromagnet

$$\textcircled{1} \quad E_N(\vec{S}) = -\frac{J}{2N} \sum_{i,j=1}^N S_i S_j \quad S_i = \pm 1 \quad N\text{-spin system}$$

$$\textcircled{2} \quad E_{N+1}(\vec{S}, S_{N+1}) = -\frac{J}{2N} \sum_{i,j=1}^{N+1} S_i S_j$$

add new spin $S_{N+1} \in \{\pm 1\}$
normalization: $\frac{1}{N} \sim \frac{1}{N+1}$ for large N

$$= -\frac{J}{2N} \sum_{i,j=1}^N S_i S_j - S_{N+1} \left(\frac{J}{N} \sum_{i=1}^N S_i \right) \\ h_N(\vec{S})$$

$$\textcircled{3} \quad E_{N+1}(\vec{S}) = E_N(\vec{S}) - S_{N+1} h_N(\vec{S})$$

$$E_{N+1}(\vec{s}, S_{N+1})$$

new spin

$$h_N(\vec{s}) = \frac{J}{N} \sum_{i=1}^N s_i$$

• so to compute the marginal of S_{N+1}

we need only consider:

$$P(S_{N+1}, \vec{s}) \rightarrow P(S_{N+1}, h_N(\vec{s}))$$

④ now we expect probability distribution of

$h_N(\vec{s})$ to concentrate around a

$$\text{deterministic value } J_m = \frac{J}{N} \sum_{i=1}^N \langle s_i \rangle$$

where

$$m = \frac{1}{N} \sum_{i=1}^N \langle s_i \rangle$$

is an unknown statistic of the coupling system

⑤ so $P(S_{N+1}, h_N(\vec{s})) \propto e^{\beta S_{N+1} h_N(\vec{s})} \delta[h_N(\vec{s}) - J_m]$

joint distribution of new spin and coupling field

so marginal $P(S_{N+1}) \propto e^{\beta J_m S_{N+1}}$

⑥ so $\langle S_{N+1} \rangle = \tanh \beta J_m$

we have computed marginal of new spin statistics

assuming a given statistic of coupling system

$$h_N(\vec{s}) \rightarrow J_m$$

$$\frac{1}{N} \sum_{i=1}^N s_i \rightarrow m$$

⑦ self consistency demands that

$$\langle S_{N+1} \rangle = m \text{ also}$$

$$\Rightarrow m = \tanh \beta J m$$

self consistent mean field equation

Next example: spin glass + ferromagnet

$$E_N = - \frac{\lambda}{2N} \sum_{i,j=1}^N s_i s_j - \frac{1}{2} \sum_{i,j} E_{ij} s_i s_j$$

ferromagnetic term

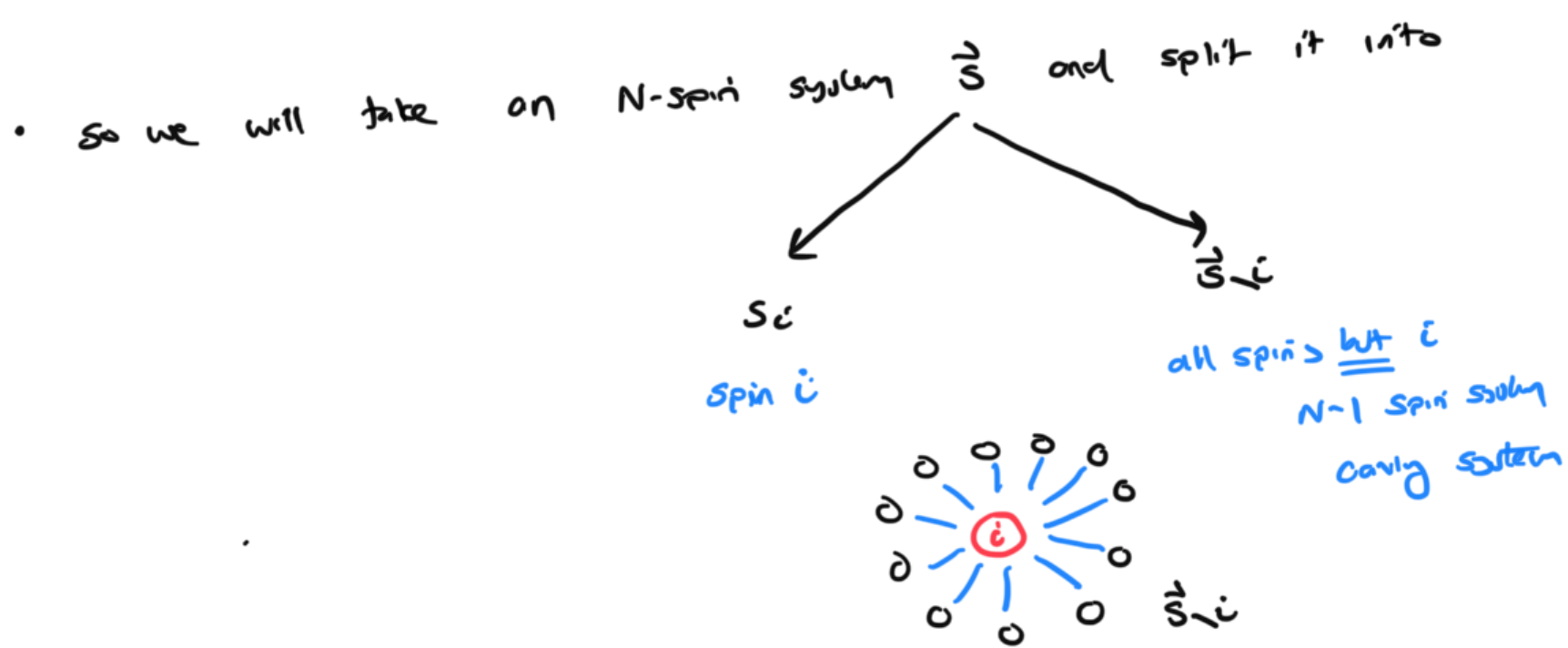
ie $\lambda = -1 \text{ or } +1$

spin glass term

$$E_{ij} \sim N(0, 1/N)$$

in comparison to previous

- now each spin will have a different magnetization $\langle S_i \rangle$ therefore it will have a different coupling field h_i



$$E_N(\vec{S}) = -\frac{\lambda}{2N} \sum_{k,l=1}^N S_k S_l - \frac{1}{2} \sum_{k,l=1}^N E_{kl} S_k S_l$$

$$= -\frac{\lambda}{2N} \sum_{k,l \in \setminus i} S_k S_l - \frac{1}{2} \sum_{k,l \in \setminus i} E_{kl} S_k S_l$$

$$- S_i \left[\lambda \frac{1}{N} \sum_{k \in \setminus i} S_k + \sum_{k \in \setminus i} E_{ik} S_k \right]$$

$$E_N(\vec{S}) = E_{-i}(\vec{S}) - S_i h_i$$

energy of coupling system of all spins but i

spin i

coupling field acting on spin i a function of $\vec{S}_{-i}$ in coupling system before spin i is added

$$h_i = \lambda \frac{1}{N} \sum_{k \in \setminus i} S_k + \sum_{k \in \setminus i} E_{ik} S_k$$

assume it converges to a deterministic quantity λM

where $M \equiv \frac{1}{N} \sum_{k \in \setminus i} \langle S_k \rangle$

M is currently unknown...

random variable $\tilde{h}_i$ in coupling system

- the coupling system of all spins $k \neq i$ does not know about the new couplings E_{ik}

- so E_{ik} independent of S_k

key usefulness of cavity method

- so $\tilde{h}_i$ should be gaussian
- its mean and variance in cavity system for fixed coupling E_{ik} is

$$\langle \tilde{h}_i \rangle = \sum_{k \in \partial i} E_{ik} \langle S_k \rangle$$

$$\text{Var}(\tilde{h}_i) = \langle \tilde{h}_i^2 \rangle - \langle \tilde{h}_i \rangle^2$$

$$\langle \tilde{h}_i^2 \rangle = \sum_{k, l \in \partial i} E_{ik} E_{il} \langle S_k S_l \rangle$$

$$\langle \tilde{h}_i \rangle^2 = \sum_{k, l \in \partial i} E_{ik} E_{il} \langle S_k \rangle \langle S_l \rangle$$

$$\text{Var}(\tilde{h}_i) = \sum_{k, l \in \partial i} E_{ik} E_{il} \text{Cov}(S_k, S_l)$$

$$\text{Cov}(S_k, S_l) = \langle S_k S_l \rangle - \langle S_k \rangle \langle S_l \rangle$$

Assumption: $\text{Cov}(S_k, S_l)$ for $k \neq l$
 $\rightarrow 0$ as $N \rightarrow \infty$

$$\approx \sum_{k \in \partial i} E_{ik}^2 \text{Var}(S_k)$$

$$= \sum_{k \in \partial i} E_{ik}^2 [1 - \langle S_k^2 \rangle]$$

$$\approx \frac{1}{N} \sum_{k \in \partial i} [1 - \langle S_k^2 \rangle]$$

$$= 1 - \underbrace{\frac{1}{N} \sum_{k \in \partial i} \langle S_k^2 \rangle}_q$$

$$\langle E_{ik}^2 \rangle = \frac{1}{N}$$

detailed realization does not matter

Summary:

$$E_N(\vec{S}_{\setminus i}, S_i) = E_{\setminus i}(\vec{S}_{\setminus i}) - h_i S_i$$

where
$$h_i = \underbrace{\lambda \left[\frac{1}{N} \sum_{k \in \partial i} S_k \right]}_m + \underbrace{\sum_{k \in \partial i} E_{ik} S_k}_{N(\langle \tilde{h}_i \rangle, 1-q)}$$

$$m = \frac{1}{N} \sum_{k \in \partial i} \langle S_k \rangle$$

$$\langle \tilde{h}_i \rangle = \sum_{k \in \partial i} E_{ik} \langle S_k \rangle$$

$$q = \frac{1}{N} \sum_{k \in \partial i} \langle S_k^2 \rangle$$

so
$$P(h_i) \sim N(\lambda m + \langle \tilde{h}_i \rangle, 1-q)$$

$\langle \cdot \rangle_{\setminus i}$ denotes an avg over the cavity system of $N-1$ spins

distribution of cavity field in $N-1$ spin system before spin i is connected

m and $\langle \tilde{h}_i \rangle$ are unknown....

then

$$P(s_i, h_i) \propto e^{\beta h_i s_i} P(h_i)$$

joint distribution of new spin s_i and coug field h_i
ofw s_i is connected to the coug

now

$$\langle s_i \rangle = \sum_{s_i = \pm 1} \int dh_i P(s_i, h_i) s_i$$

$$\langle s_i \rangle = \tanh \beta [\lambda_M + \langle \tilde{h}_i \rangle]$$

we have now computed the marginal of the new spin s_i in terms of the statistics of the coug field only its unknown mean $\lambda_M + \langle \tilde{h}_i \rangle$ happens to show up....

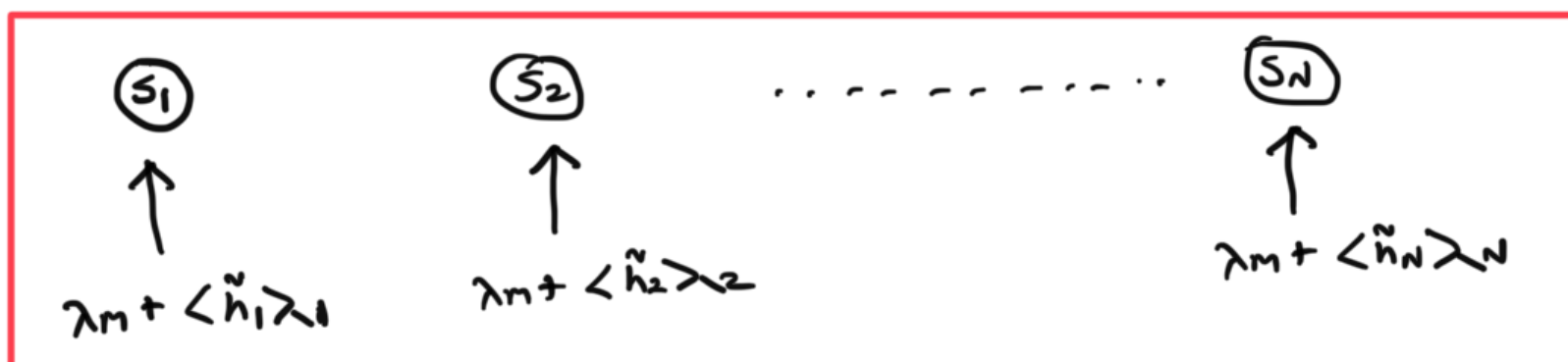
Now to obtain self-consistent equations for q and M

- Note we could have split apart $\vec{S} \rightarrow \begin{matrix} \vec{S}_i \\ s_i \end{matrix}$ for any i

- each split would have an associated

equation:

$$\langle s_i \rangle = \tanh \beta [\lambda_M + \langle \tilde{h}_i \rangle]$$



- Split up connected system into a set of independent systems where each spin i is driven by a different coug field $\lambda_M + \langle \tilde{h}_i \rangle$

- can derive self consistent equations for q and M by avg over all N spins

- first: what is the statistics of $\langle \tilde{h}_i \rangle$ across index i
 at large N the histogram of $\langle \tilde{h}_i \rangle$ converges to Gaussian distribution

$$\langle \langle \tilde{h}_i \rangle \rangle_E = \left\langle \sum_{k \in i} E_{ik} \langle s_k \rangle \right\rangle_E = 0$$

$$\begin{aligned} \langle [\langle \tilde{h}_i \rangle]^2 \rangle_E &= \left\langle \sum_{k, l \in i} E_{ik} E_{il} \langle s_k \rangle \langle s_l \rangle \right\rangle_E \\ &= \sum_{k, l \in i} \underbrace{\langle E_{ik} E_{il} \rangle_E}_{\text{blue underline}} \langle s_k \rangle \langle s_l \rangle \end{aligned}$$

$$\frac{1}{N} \sum_{i=1}^N s_i^2$$

$$= \sum_{i=1}^N \frac{1}{N} \langle s_i^2 \rangle = q$$

Summary: the histogram of couplings across the site index i converges as $N \rightarrow \infty$ to the gaussian distribution $\mathcal{N}(0, q)$

now assume m and q are the same in the $N-1$ system and the full N system:

$$m = \frac{1}{N} \sum_{i=1}^N \langle s_i \rangle$$

$$= \frac{1}{N} \sum_{i=1}^N \tanh \beta [\lambda m + \langle \tilde{h}_i \rangle_i]$$

$$m = \int \mathcal{D}z \tanh \beta [\lambda m + \sqrt{q} z]$$

recovers first replica equation !!!

$$\xrightarrow{N \rightarrow \infty} \int \mathcal{D}z \tanh \beta [\lambda m + \sqrt{q} z]$$

and

$$q = \frac{1}{N} \sum_{i=1}^N \langle s_i^2 \rangle$$

$$= \frac{1}{N} \sum_{i=1}^N \left(\tanh \beta [\lambda m + \langle \tilde{h}_i \rangle_i] \right)^2$$

$$q = \int \mathcal{D}z \tanh^2 \beta [\lambda m + \sqrt{q} z]$$

recovers second replica equation !!!

$$\xrightarrow{N \rightarrow \infty} \int \mathcal{D}z \tanh^2 \beta [\lambda m + \sqrt{q} z]$$